Comenius University in Bratislava
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Browser Automation with Natural Language Instructions Utilizing Multimodal Methods

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I hereby declare that I have written this master thesis by myself, only with help of referenced literature and tools of artificial intelligence, under the careful guidance of my thesis supervisor. I have used artificial intelligence tools in accordance with legal regulations, academic guidelines for the use of artificial intelligence at Comenius University, ethical and moral principles, while maintaining academic integrity, and that their use is appropriately identified in the work.

During the preparation of this work, the UI tools DeepL and Claude were used for the purpose of translation and stylistic edits and grammar checking. The author is responsible for the accuracy of the resulting text.

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Abstract

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Chapter 1

Introduction

Agentic pipelines make use of large language models to interact with the environment via APIs or graphical interfaces. Pivot to agentic pipelines is however plagued by multiple problems. An error in a single step propagates further, requiring methodological regulations to be reliable and accurate. Complex pipelines are therefore facilitated by a large amount of computational power. While APIs are highly preferred as means for LLM agent interactions, they are not accessed easily, as the primary application consumer has until now had been a human utilizing a graphical interface.

Agentic systems utilize grounding methods along with a planning model, usually an LLM, to plan and execute wide array of long-term tasks. An agent interacting with a web could book a ticket, buy a product from an e-shop, search for the most recent information on the news or access social media. (Zheng et al., 2024).

1.1 Planning model

The planning model decomposes user instruction into atomic steps and adapts to changes in the environment. The recent advances introduce a reflection that helps the agent recover from failures, encourage short-term learning and avoid hallucinations. Summarization serves to distill information from each step into comprehensible output. (Guo et al., 2026)

1.2 Grounding method

The LLMs are inherently based on natural language concepts. The agentic process requires the system to operate within the constraints of the environment and map the the fuzzy natural language to specific elements or executable actions. Such mapping requires environmental grounding - the linking of concepts to context. (Chandu and Bisk and Black, 2021)

Zheng et al. (2024) show that multimodal GPT-4V as a planner is able to execute wide range of tasks across multiple domains given a hypothetical "oracle" UI grounding method. The "oracle" is approximated by human annotators who identify and ground the model's intentions. Along the grounding method an in-context examples were provided. This approach achieves up to 65% step accuracy utilizing cross-website in-context examples. In comparison, a multimodal grounding method `SeeAct`, based on the works of Deng et al. (2023), achieves 32.7% accuracy. This experiment highlights the importance of grounding in web agents.

In the case of web navigation, the grounding is applied in two interconnected layers - UI grounding and action grounding. The LLM is contextualized within the environment composed of elements such as sections, headers, paragraphs or buttons. Each of those elements carries an information about the environment and acts as context for the concepts LLM operates with. The contextualized actions provide available operations within environment that can be taken at each step. (Deng et al., 2023)

1.3 Previous works

One of the first specialized approaches to GUI grounding, `MindAct`, entails filtering the elements of the web page using smaller language model. Elements are scored based on semantic similarity using an encoder-only model. The top-k elements with the highest scores are provided to the LLM as a multi-choice question answering problem. The elements are clustered into smaller subsets which are fed into the LLM along with a "None" option, one subset at a time. If the LLM selects multiple elements, new subsets are created out of the selected elements. The process repeats until the last element remains or LLM chooses "None" for all subsets. (Deng et al., 2023)

Aiming to generalize across multiple platforms such as Windows applications, websites and mobile devices, the agentic workflows pivot to using purely visual information - screenshots - for UI grounding. In `UGround` the LLM operates on pixel-level actions that are contextualized using the grounding model. The model is based on LLaVA-1.5 but during instruction post-training uses the `AnyRes` technique to allow for variable image resolution on input. Then the model is trained on a synthetic dataset of 9M web elements. (Gou et al., 2025)

1.4 Problems of agentic workflows

While the `UGround` is able to complete array of tasks across multiple domains, as outlined in Xue et al. (2025), the development in the field of agents as a whole faces multiple issues. Advances are usually presented in the context of various benchmarks,

which are usually based on task accuracy. This does not represent the usefulness of the grounding method in practice as the task grounding may fail multiple times while the task as a whole is ultimately successful. Existing methods are still distant from being useful on local devices as benchmarks do not prioritize or report the number of steps and time taken needed to accomplish the tasks. Many models are not optimized to fit on commercial devices. Moreover, benchmarks are skewed and do not capture the whole dynamics of the web environment, focusing mostly on e-commerce or booking services. Large amount of traffic is concentrated on dynamic websites like social media and text heavy platforms such as news or forums.

1.5 Thesis goals

We make case for an approach to navigating a web browser application, that manages to address multiple issues while utilizing already existing graphical interface in a low-cost manner. Such approach is formulated into multiple tasks:

1. Explore and extend web agent benchmarks with performance metrics from existing methods. Diversify benchmarks with samples from different domains.

2. Design and implement more efficient approach to multimodal environment representation in the context of web agents. Verify the novel approach on existing benchmarks.

Chapter 2

Methodology

We present approach to an effective grounding during LLM based browser automation. This method operates on a DOM tree and screenshot of a rendered website and evaluates a user prompt. Based on natural language instructions the method determines an appropriate action and a grounded element. As an example, the instruction "Open shopping cart" evaluates into "action: click" and "element id: 14", where element with ID 14 maps into a clickable icon that opens a shopping cart menu; or instruction "Scroll video" evaluates into "action: scroll from top" and "element id: 1", where element with ID 1 maps into a video reel. The output is designed to be easily parsed by any browser automation tool such as Selenium or Playwright.

Our method is separated into two steps. In accordance with Deng et al. (2023), the steps are referred to as *Candidate Generation* and *Action Prediction*. The former operates on multimodal representation of a website, namely textual and visual representation. The interleaved representations are pruned and elements that convey the most information are selected. These representative elements, called candidates, are along with their position processed in the *Action Prediction* step. The candidates are embedded with their position, either in text-first approach or image-first approach. Candidates are processed by a large language model to determine the appropriate action and element from the pool of candidates in a multi-choice manner.

2.1 Candidate Generation

During the first step, the website is transformed, aiming to reduce the amount of data, while preserving the information contained in those data. This *Candidate Generation* process is essential due to the large amount of data representing each website and facilitates more advanced methods that would otherwise be too expensive to use.

2.1.1 Multimodal input data

The website environment is represented using multimodal data comprising textual and visual representations. The **textual representation** is naturally available from browser as a Document Model Object (DOM) where each node represents a rendered element. While the HTML contains initial structure, DOM is more suitable as it describes current state of the document. These data are relatively inexpensive to obtain but their hierarchical structure is first and foremost intended for a compiler and does not convey a semantic representation. For example, elements that are grouped together in a single section might be rendered at different location or be absent entirely based on the size of a display, type of device or a JavaScript function call. Large portions of the data consist of element attributes, CSS styles or JavaScript snippets, that are many times difficult to follow or do not convey a meaningful information. However, all textual representations depend on correct coding practices and are unable to capture images or videos embedded in the website.

The **visual representation**, i.e., screenshot of the website, is intended for a human audience and therefore should naturally aim to capture the semantics of the website. This representation conveys spatial relations between elements on a two-dimensional plane instead of a hierarchical tree. The downside of the visual approach is its cost, processing a high fidelity screenshot requires substantially more resources than a structured DOM.

Interwoven multimodal data provide ability to process text elements and estimate their importance in a low-cost manner, encompass the semantics of the environment and relate elements rendered adjacently.

2.1.2 Element pruning

The rendered website usually contains large amount of elements. It is computationally ineffective to process this data using an LLM directly. Therefore we remove nodes with low informational value:

1. DOM is pruned of all metadata nodes such as `head` or `metadata`
2. Node attributes with values that are not meaningful are removed
3. Nodes without any attributes are removed

2.1.3 Inter-modal mapping

The DOM nodes include information about the element's position on a rendered viewscreen. This information is used to extract a slice from the screenshot that matches the element rendered from the attribute. The mapping produces a set of elements with

information about their DOM attributes and rendered appearance. The mapping may cause the duplication of information therefore if the text takes up at least XY% of the element, the visual information is removed.

2.1.4 Visual-based element detection

To complement the mapped element set, additional detection is needed. For example, more complex interactive elements like interactive maps or captchas are not represented by a DOM tree and require further transformation. A lightweight visual model, e.g. of YOLO family, finetuned to detect UI elements is applied to a screenshot. Resulting bounding boxes are either merged with existing elements or used to create new elements.

2.2 Intermediate Candidate Elements

In the previous step we applied both text-based and vision-based recognition to detect elements on the website. We also removed elements deemed as meaningless. The resulting pool of elements, also called candidates, serves as a sparse representation of the website. Each candidate comprises textual and/or visual representation along with its position on two dimensional grid. The visual features are computationally more expensive to process compared to textual by a multimodal LLM. Therefore we aimed to maximize the portion of textual data in comparison to images. The size of the set of candidates can range from few dozens to hundreds of candidates, scaling based on the complexity of the website. The segmentation of the website helps LLM understand the structure of the environment and decrease computational requirements while preserving the information value.

2.3 Action Prediction

The role of candidate elements in action prediction is twofold. First, the candidates represent the website in a structured manner. Second, individual candidates can be interacted with, grounding the model during the navigation step. Specifically, the task is formulated as a multi-choice decision between the candidates given a natural language instruction. A multimodal LLM is used to select a candidate from the pool along with a specific action. Before the candidates are passed into the model, the candidates undergo transformation, where their position is embedded. Transformation helps us represent the candidates' positions in such manner that is more natural to the multimodal LLMs. Our approach proposes two different transformation methods:

2.3.1 Text-first positional transformation

Transformation leverages that markup language documents (such as HTML) are included in training data of majority of large language models. The pool of candidates is parsed into a simplified HTML data. This data are structured hierarchically from textual representations of the candidates. Visual representations are marked with a unique identified which is included in the textual representation, linking visual and textual representations of the same element together. Simplified HTML and visual representations are processed by the multimodal LLM along with the instruction.

2.3.2 Image-first positional transformation

Transformation employs positional embedding that embeds two dimensional position directly into candidates' token representation. The embedding is based on 3D-RoPE Ma et al. (2025) that facilitates positional embedding of text on a two dimensional grid representing the website. This provides a natural way to represent positions and proximity between elements. The disadvantage is the transformation requires a fine-tuning that adapts decoder to the introduced embeddings.

2.4 LLM-based Navigation

The multi-modal LLM is instructed by a prompt. The prompt formulates the navigation as a multi-choice problem, given available actions and list of transformed candidates. Model is directed to select a specific action that should be executed to complete the user's request. Specific descriptions of available actions are included:

- Actions `left click` and `right click` should include a single element where the click will be executed.
- Action `scroll` should provide an element which will be scrolled, direction of the scroll and length of the scroll.
- Action `drag` should include two elements - origin position and goal position of the drag.
- Action `type` should include an element which will be focused and a text to be typed.

The model is then supplied with the user instruction and transformed candidates. The instructions are limited to be completed in a single step. The output of the model is then parsed as the action to be executed in the browser.

Chapter 3

Exploration of existing approaches

- Prabhu et al. (2026) (text) **WALT** Grounding by generating and iterating tools.
- Team et al. (2026) (visual, text) **UI-Venus 1.5** An end-to-end agentic framework including dense models and MoE. Trained using reinforcement learning.
- Wang et al. (2026) (visual, text) **ColorBrowserAgent** A framework helping long-horizon stability of an LLM Agent. Human in the loop, comprehensive prompt engineering.
- Guo et al. (2026) (visual, text) **OpAgent** A method for comprehensive web navigation, using planning, grounding, reflecting and summarizing. For grounding a fine-tuned Qwen2.5-VL-7B is used.
- Xie et al. (2026) (benchmark, dataset) **OSWorld-G** Benchmark for grounding and interacting with operating system, applications, web... Creates a **JEDI** dataset for grounding that is used to fine-tune multiple multimodal LLMs.
- Gou et al. (2025) (visual, dataset) **UGround** Purely visual model based on LLaVA for grounding UI on multiple platforms and applications. Dataset containing 10M GUI elements over 1.3M screenshots. Utilizes **SeeAct-V** (visual-only adaptation of SeeAct framework).
- Krupp et al. (2025) (survey, benchmark) Very nice survey of web agents and their benchmarks, identifying lack of comparability and high computational efficiency.
- Yang et al. (2025) (visual) **Aria-UI** Trained grounding mixture-of-expert model on web, desktop and mobile environments. Visual-only, using large amount of data from Common Crawl collection and LLM based annotations, with visual, positional and functional captioning.

- Xue et al. (2025) (survey, benchmark) **Online-Mind2Web** benchmark for web agents and **WebJudge** LLM-as-a-judge for an automatic evaluation of web agent tasks.
- Verma et al. (2025) (visual, text) **AdaptAgent** A framework with few-shot human demonstrations, uses SeeAct prompt in combination with proprietary model GPT-4o and open-weights model CogAgent. Uses meta-training instead of finetuning to improve model performance. Benchmarked on Mind2Web and VisualWebArena.
- Song et al. (2025) (benchmark) Benchmark **BearCubs** with text-based and multimodal complex interactions.
- Nguyen et al. (2025) (survey) 2024 Survey of web agents, benchmarks, evaluation metrics.
- Erdogan et al. (2025) (text, visual) **Plan-And-Act** splits into planner and executor, executor grounds GUI, is implemented by a LLaMA-3.3-70B-Instruct finetuned on synthetic data.
- Zheng et al. (2024) (text, visual) **SeeAct** Explores grounding via element attributes (generating detailed attribute closest to the target element followed by heuristic search in HTML), textual choices (rank and choose top-k candidates which are presented as a multi-choice question), image annotations (overlay a bounding box for each candidate element selected by the ranker), and a "manual" oracle action grounding (all details to identify elements are in the action description). The oracle action grounding is deemed as the best, textual choice grounding using GPT-4V is ranked as second.
- Z. Zhang and A. Zhang (2024) (visual) **Auto-GUI** Encoder for screenshots combined with action history fed into LLM.
- Xu et al. (2024) (visual) **Aguvis-7B** Custom model trained on screenshots. Based on Qwen2-VL with 2D-RoPE embeddings.
- Tao et al. (2024) (text) **WebWISE** Using filtered DOM tree as an LLM input.
- Lutz et al. (2024) (text) **WILBUR** In context demonstration generation and retrieval.
- Lu et al. (2025) (text, visual) **Omniparser** Has separate model to detect interactable elements using Set-of-Mark and OCR model to detect text. An image with bounding boxes with box description fed into LLM.

- He et al. (2024) (visual) **WebVoyager** Uses Set-of-Mark to mark interactable elements, runs VLLM (Claude/GPT-4o) on the resulting image. Evaluated automatically using GPT-4V and on benchmarks GAIA and SeeAct.
- Cheng et al. (2024) (visual, benchmark) **Seeclick** Android and web navigation using custom model trained on UI grounding data. Introduces grounding benchmark **ScreenSpot**.
- Furuta et al. (2024) (text, visual) **WebGUM** Encoder-decoder processing screenshots, instructions and HTML. ImageNet-21K for vision transformer. Evaluated on MiniWoB++ and WebShop benchmarks.
- Deng et al. (2023) (text, dataset) **Mind2Web** dataset for evaluating web agents. **MindAct** a small model for candidate generation from elements present on the webpage, selection using large LLM to predict the final action.

3.1 Comparison of UI Grounding methods

Method	Text	Vision	Approach
MindAct	✗	✓	DeBERTa to choose top-k candidates → multi-choice
$\text{SeeAct}_{\text{Choice}}$	✓	✓	multimodal MindAct
$\text{SeeAct}_{\text{Annotation}}$	✓	✓	Set-of-Mark (mark interactable elements)
OpAgent	✓	✓	finetuned Qwen2.5-VL-7B
UGround	✗	✓	custom LLaVA model
Aria-UI	✗	✓	25B visual mixture-of-expert grounding model
AdaptAgent	✓	✓	few-shot demonstrations or meta-learning
Plan-And-Act	✓	✓	finetuned LLaMA-3.3-70B-Instruct
Auto-GUI	✗	✓	visual encoder
Aguvis-7B	✗	✓	custom visual model
WebWISE	✓	✗	filtered DOM
WILBUR	✓	✗	In context examples
Omniparser	✓	✓	visual model for Set-of-Mark + OCR
WebVoyager	✗	✓	Set-of-Mark using GPT-4V-ACT
Seeclick	✗	✓	custom visual model
WebGUM	✓	✓	encoder-decoder, based on ImageNet-21K
UI-Venus	✗	✓	MoE model trained using reinforcement learning
WALT	✓	✗	Generating and iterating tools

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